Adaptive Decision Intelligence for Crisis Management

A Classical AI-Based Decision Support System Combining Rule-Based Reasoning, Heuristic Search, and Constraint Satisfaction for Emergency Response Planning

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Abstract - Disaster and crisis situations such as floods, earthquakes, cyclones, and wildfires require rapid, well-coordinated decisions under conditions that change quickly and involve many interacting factors. This paper presents Adaptive Decision Intelligence for Crisis Management, a classical Artificial Intelligence-based decision support system that assists emergency authorities in analyzing a crisis and selecting a feasible response strategy without relying on machine learning or deep learning. The system combines rule-based reasoning over a disaster- and severity-specific knowledge base, heuristic search for comparing alternative response strategies, and Constraint Satisfaction Problem (CSP) techniques for validating resource feasibility. An adaptive decision engine re-evaluates the recommendation whenever new information becomes available, allowing the system to track a dynamically evolving emergency rather than produce a single fixed output. A Streamlit-based interface allows crisis parameters to be entered and recommendations to be viewed. Results show that the system can generate, score, and validate multiple response strategies and revise its recommendation as conditions change, demonstrating that fundamental AI techniques—logical reasoning, search, and constraint satisfaction—can be combined to build an explainable and practically feasible decision-support tool for emergency management.

Keywords—crisis management, decision support system, rule-based reasoning, heuristic search, constraint satisfaction problem, adaptive decision-making, artificial intelligence

I. INTRODUCTION

Disaster and crisis situations such as floods, earthquakes, cyclones, and wildfires require immediate and well-coordinated decisions. During such emergencies, authorities must simultaneously consider several factors including the severity of the disaster, affected regions, weather conditions, availability of rescue teams, medical facilities, transportation resources, and shelter capacity. Since these conditions can change rapidly, making decisions manually can result in delays, inefficient resource utilization, and difficulty in selecting the most appropriate response strategy. Artificial Intelligence provides a suitable approach for supporting decision-making in such complex environments through reasoning, search, and constraint-solving techniques.

Adaptive Decision Intelligence for Crisis Management is an AI-based decision support system designed to assist emergency authorities in analyzing crisis situations and selecting suitable response strategies. The system uses classical Artificial Intelligence concepts rather than machine learning or deep learning. It combines knowledge representation, rule-based reasoning, heuristic search, and Constraint Satisfaction Problem techniques to analyze emergency conditions and generate feasible recommendations.

The system begins by collecting information about the current crisis. The input includes the type of disaster, severity level, affected region, weather condition, number of rescue teams, medical facilities, transportation vehicles, and shelter capacity. This information is processed by a knowledge base containing disaster-specific and severity-specific rules. The rule-based reasoning engine evaluates these conditions and determines the priority and urgency of the situation.

The system then generates multiple possible response strategies. A heuristic search mechanism evaluates these alternatives using estimated response time, resource utilization, and operational risk. The most promising strategy is identified based on its heuristic cost. However, a strategy that appears efficient may not always be practically executable; therefore the CSP module verifies whether sufficient rescue teams, medical facilities, transportation resources, and shelter capacity are available.

A major feature of the system is its adaptive decision engine, which re-evaluates the crisis and generates an updated recommendation whenever new information becomes available, allowing the decision-support system to respond to changing emergency conditions rather than producing a fixed recommendation.

II. PROBLEM STATEMENT AND MOTIVATION

A. Problem Statement

During emergency situations, decision-makers must process large amounts of information within a very limited time. Traditional crisis management approaches depend heavily on manual assessment and coordination between agencies. When several factors change simultaneously, it becomes difficult to identify the best course of action quickly.

For example, during a cyclone, an emergency authority may need to determine which areas should be evacuated first, how many rescue teams should be deployed, which hospitals should receive victims, how many vehicles should be assigned for evacuation, and whether available shelters can accommodate the affected population. A decision that is appropriate at one point may become unsuitable when weather conditions worsen or resources become unavailable.

The problem addressed by this project is therefore to design an intelligent decision-support system capable of analyzing changing crisis conditions and recommending an appropriate response strategy while considering practical resource constraints.

B. Motivation

The primary motivation for developing this system is the need for faster, systematic, and explainable decision-making during emergencies. Rule-based reasoning can represent expert knowledge using logical conditions—for example, that a cyclone of critical severity under extreme weather should trigger immediate evacuation—allowing the system to provide understandable reasoning rather than an unexplained result.

Heuristic search is useful because a crisis may have several possible response strategies; instead of selecting an action arbitrarily, the system evaluates alternatives based on estimated response time, resource usage, and risk. CSP provides a further safeguard by ensuring that recommendations satisfy real-world limitations: a strategy requiring five rescue teams cannot be executed if only two are available, and a shelter cannot be assigned more people than its capacity.

The adaptive component is motivated by the dynamic nature of disasters, whose conditions can change rapidly due to weather, resource availability, or increasing severity. The project demonstrates how fundamental AI concepts can be integrated into a practical application for emergency decision support.

III. OBJECTIVES

The main objective of Adaptive Decision Intelligence for Crisis Management is to develop a classical AI-based decision-support system that can analyze crisis situations and recommend effective and feasible emergency response strategies. The specific objectives are:

- To develop an intelligent crisis assessment module that accepts key emergency parameters such as disaster type, severity, weather conditions, available resources, and shelter capacity.
- To create a knowledge base containing disaster-specific, severity-specific, and resource-related information required for decision-making.
- To implement rule-based reasoning for evaluating emergency conditions and identifying priority and urgency.
- To generate multiple response strategies such as evacuation-first, rescue-first, medical-first, and balanced emergency strategies.
- To implement heuristic evaluation for comparing alternative strategies based on response time, resource usage, and operational risk.
- To implement Constraint Satisfaction Problem techniques for verifying whether a selected strategy can be executed using available resources.
- To prevent resource conflicts by ensuring recommendations do not exceed available rescue teams, vehicles, medical facilities, or shelter capacity.
- To develop an adaptive decision mechanism that re-evaluates recommendations whenever new crisis information is provided.
- To provide explainable recommendations so users can understand why a strategy was selected.
- To integrate classical AI techniques into a single practical application demonstrating FOAI concepts.
- To improve emergency-planning efficiency by reducing the time required to compare response strategies.
- To develop a user-friendly interface for entering emergency information and viewing AI-generated recommendations.

The overall goal is not to replace human emergency authorities but to provide them with an intelligent supporting system that processes crisis information systematically and assists faster, more reliable decision-making.

IV. EXISTING SYSTEM AND PROPOSED SYSTEM

A. Existing System

Traditional disaster management systems generally depend on predefined procedures, manual assessment, inter-departmental communication, and human decision-making. Although human expertise is essential, manual decision-making becomes difficult as the number of variables increases, since a plan created initially may become unsuitable as weather worsens or resources decrease. Comparing multiple response strategies manually can also consume valuable time, and resource conflicts may occur when several operations require the same resources.

B. Limitations of Existing Approaches

- High dependence on manual decision-making.
- Difficulty processing multiple crisis parameters simultaneously.
- Delays in comparing alternative response strategies.
- Possibility of inefficient resource allocation.
- Difficulty adapting quickly to changing conditions.
- Lack of systematic constraint validation.
- Difficulty maintaining consistency across decisions.

C. Proposed System

The proposed system addresses these limitations using classical AI techniques, providing a structured process for collecting crisis information, evaluating the situation, generating response strategies, comparing them using heuristic reasoning, and validating the selected strategy using constraints. Crisis information is received through a user interface; a knowledge base stores disaster and severity rules; the rule engine produces an initial assessment; a heuristic search module generates alternative strategies and calculates a score for each; and a CSP module validates feasibility against available resources, rejecting a strategy and considering an alternative when required resources exceed availability. An adaptive decision engine repeats this process when updated information is received, making the system suitable for dynamic crisis scenarios.

D. Advantages of the Proposed System

- Faster analysis of crisis situations.
- Systematic, explainable decision-making.
- Better utilization of available resources.
- Explicit constraint validation.
- Adaptive response to changing conditions.
- Use of classical, interpretable AI techniques.
- User-friendly decision-support interface.

V. SYSTEM ARCHITECTURE AND AI METHODOLOGY

The proposed system follows a sequential AI decision-making architecture in which the output of one stage becomes the input for the next:

Crisis Input → Knowledge Base → Rule-Based Reasoning → Response Plan Generation → Heuristic Search → CSP Validation → Adaptive Re-evaluation → Final Recommendation

1) Crisis Input:

The user provides the disaster type, severity, affected region, weather condition, rescue teams, medical facilities, vehicles, and shelter capacity.

2) Knowledge Base:

Stores disaster-specific rules describing the primary response required for floods, earthquakes, cyclones, and wildfires, together with severity rules that determine response priority and urgency.

3) Rule-Based Reasoning:

The rule engine compares the current situation with predefined logical conditions to identify emergency priorities, risks, and recommended actions, for example:

IF disaster = Cyclone AND severity = High THEN activate emergency response and prioritize evacuation.

4) Response Plan Generation:

The system generates alternative strategies—evacuation-first, rescue-first, medical-first, and balanced—rather than directly selecting one action.

5) Heuristic Search:

Each strategy receives a heuristic score based on response time, resource usage, and risk. The heuristic function used is:

$$H(n) = \text{Response Time} + \text{Resource Usage} + \text{Risk Level} \quad (1)$$

A lower heuristic cost indicates a more efficient strategy.

6) CSP Validation:

The selected strategy is checked against available resources, ensuring requirements do not exceed the available rescue teams, medical facilities, vehicles, and shelter capacity.

7) Adaptive Decision Engine:

When new information is introduced, the system compares the initial and updated situations and repeats the reasoning, search, and constraint-validation process.

8) Final Recommendation:

The final output provides the recommended strategy, heuristic score, resource requirements, constraint status, warnings, and recommended emergency actions.

VI. MODULE DESCRIPTIONS

A. Module 1: Crisis Input and Situation Assessment

This module collects the information required for the AI system to understand the current crisis. Accurate input is essential because subsequent reasoning depends on these parameters: disaster type (flood, earthquake, cyclone, or wildfire), severity level (low, moderate, high, or critical), affected region, weather condition (clear, rain, heavy rain, storm, or extreme), and resource counts for rescue teams, medical facilities, vehicles, and shelter capacity.

The module is implemented using a Streamlit-based interface, with dropdown menus for categorical information and numerical fields for resource values. After entering the information, the user selects "Assess Crisis," and the system records and passes it to the reasoning modules. For example, a cyclone scenario of high severity, heavy rain, 3 rescue teams, 2 medical facilities, 5 vehicles, and 500 shelter capacity forms the initial state passed to the knowledge base.

B. Module 2: Knowledge Base and Rule-Based Reasoning

This module implements knowledge representation and rule-based reasoning. The knowledge base associates each disaster type with its primary response (e.g., cyclones with emergency evacuation and rescue, floods with evacuation and flood rescue), and severity rules associate each severity level with a priority and response level, as summarized in Table I.

TABLE I
Severity-Priority Mapping

Severity	Priority	Response Level
Low	Low	Routine
Moderate	Medium	Elevated
High	High	Urgent
Critical	Critical	Immediate

The rule engine evaluates conditions such as: if severity is critical, prioritize life-saving operations; if severity is high, deploy emergency response teams; if weather presents increased risk, prioritize safe transportation routes; if rescue teams are limited, prioritize critical affected zones; if medical facilities are limited, flag medical resource constraints; and if shelter capacity is critically limited, generate a shelter-capacity warning. The module outputs the emergency priority, response level, primary response action, AI recommendations, detected warnings, and resource priority, giving the user an explainable link between input conditions and resulting recommendations.

C. Module 3: Heuristic Search and Response Strategy Generation

This module applies heuristic problem-solving to compare candidate response strategies rather than selecting one immediately. Four strategies are generated: evacuation-first (identifies high-risk zones, deploys transport, evacuates people to shelters), rescue-first (prioritizes critical zones, deploys rescue teams, transports victims to medical facilities), medical-first (prepares medical facilities and resources to receive affected individuals), and balanced (combines rescue, evacuation, medical preparation, transportation, and shelter allocation).

Each strategy is scored on response time, resource usage, and risk level using H(n) as defined in (1); a lower score indicates greater efficiency. Table II shows a representative scoring example.

TABLE II
Sample Heuristic Scores for Candidate Strategies

Strategy	Time	Resources	Risk	Score
Evacuation-First	30	30	15	75
Rescue-First	30	25	20	75
Medical-First	35	20	25	80
Balanced	25	30	15	70

In this example the Balanced strategy has the lowest heuristic cost and is preferred before constraint validation. If crisis conditions change, the calculated values change too, which can shift the preferred strategy—forming the bridge between rule-based assessment and CSP-based feasibility checking.

D. Module 4: Constraint Satisfaction and Resource Validation

This module ensures the selected strategy can actually be executed with the resources available, modeled as a Constraint Satisfaction Problem with variables (rescue teams, medical facilities, transportation vehicles, shelter capacity), domains (the possible values each resource can take, e.g., zero to three teams if three are available), and constraints (required resources of each type must not exceed the available amount).

For example, an evacuation-first strategy requiring 1 rescue team, 1 medical facility, 3 vehicles, and 200 shelter capacity is feasible when 3 teams, 2 facilities, 5 vehicles, and 500 shelter spaces are available, but would fail the transportation and shelter constraints if only one vehicle and 50 shelter spaces existed. The CSP module evaluates all generated strategies, removes those that violate constraints, and among the remaining feasible plans selects the one with the lowest heuristic score—so the final decision reflects both efficiency (heuristic search) and feasibility (CSP validation), preventing the system from recommending an efficient but non-executable strategy.

E. Module 5: Adaptive Decision Engine

The Adaptive Decision Engine updates the recommendation as new information becomes available—the feature that gives the project its adaptive nature. Weather may worsen, severity may increase, or rescue teams, vehicles, medical facilities, and shelter capacity may become reduced, so a decision generated earlier may no longer be suitable.

The engine evaluates the initial state through Rule-Based Reasoning → Heuristic Search → CSP Validation, and when updated information is entered, executes the same process again, comparing the initial and updated states to detect changes in severity, weather, or resource availability. For example, an initial cyclone scenario of high severity may later be updated to critical severity with extreme weather and sharply reduced resources; the adaptive engine detects the change and re-runs the full decision cycle: Initial Crisis State → AI Assessment → Recommended Strategy → New Information → Detect Changes → Re-evaluate Rules → Regenerate Strategies → Recalculate Heuristic Values → Validate Constraints → Updated Recommendation. This prevents the system from relying on outdated information and demonstrates how classical AI techniques can be combined to create a dynamic decision-making system without machine learning or neural networks.

VII. RESULTS AND DISCUSSION

The developed system successfully integrates multiple classical AI concepts into a single crisis-management decision-support application. Given emergency information, the knowledge base and rule engine identify the urgency and primary response requirements; the heuristic search module generates and scores alternative response strategies based on response time, resource usage, and risk, enabling identification of an efficient strategy under current conditions.

The CSP module further improves decision quality by checking resource constraints, identifying strategies that require resources beyond available capacity as infeasible, so the system selects a strategy that is not only efficient but also

executable. The adaptive engine enables the system to respond to changes in severity, weather, or resource availability by re-running the decision process and generating a new recommendation.

VIII. LEARNING OUTCOMES

Through this project the following concepts were understood and implemented:

- Knowledge representation using structured disaster information.
- Rule-based reasoning using IF–THEN conditions.
- Problem-solving through alternative response strategies.
- Heuristic evaluation for comparing possible solutions.
- Constraint Satisfaction Problems for resource validation.
- State-based representation of changing crisis conditions.
- Adaptive decision-making using repeated AI evaluation.
- Integration of multiple classical AI concepts.
- Development of a practical AI application using Python.
- Design of a user-friendly interface using Streamlit.
- Understanding the importance of explainable decision-making.
- Application of theoretical FOAI concepts to a real-world problem.

IX. CONCLUSION

Adaptive Decision Intelligence for Crisis Management demonstrates how classical Artificial Intelligence techniques can be applied to support emergency decision-making. The system combines rule-based reasoning, heuristic search, and Constraint Satisfaction Problem techniques to analyze crisis conditions, compare response strategies, validate resource availability, and generate feasible recommendations. The adaptive decision engine further enhances the system by allowing decisions to be recalculated when new information becomes available, which is particularly important because emergency conditions can change rapidly.

The project shows that intelligent decision support does not necessarily require machine learning or deep learning: fundamental AI concepts such as logical reasoning, search, knowledge representation, and CSP can be effectively combined to solve complex decision-making problems. Although designed as an academic prototype rather than a replacement for professional emergency-management systems, it demonstrates the potential of classical Artificial Intelligence to assist authorities in making faster, more consistent, and resource-aware decisions during complex emergencies.

References

[1] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Boston, MA: Pearson, 2020.

[2] J. Pearl, Heuristics: Intelligent Search Strategies for Computer Problem Solving. Reading, MA: Addison-Wesley, 1984.

[3] E. Tsang, Foundations of Constraint Satisfaction. London, U.K.: Academic Press, 1993.

[4] R. Kruse, C. Borgelt, C. Braune, S. Mostaghim, and M. Steinbrecher, Computational Intelligence: A Methodological Introduction, 2nd ed. London, U.K.: Springer, 2016.

[5] United Nations Office for Disaster Risk Reduction, "Global Assessment Report on Disaster Risk Reduction," UNDRR, Geneva, Switzerland, 2023.

[6] Streamlit Inc., "Streamlit documentation," 2024. [Online]. Available: https://docs.streamlit.io